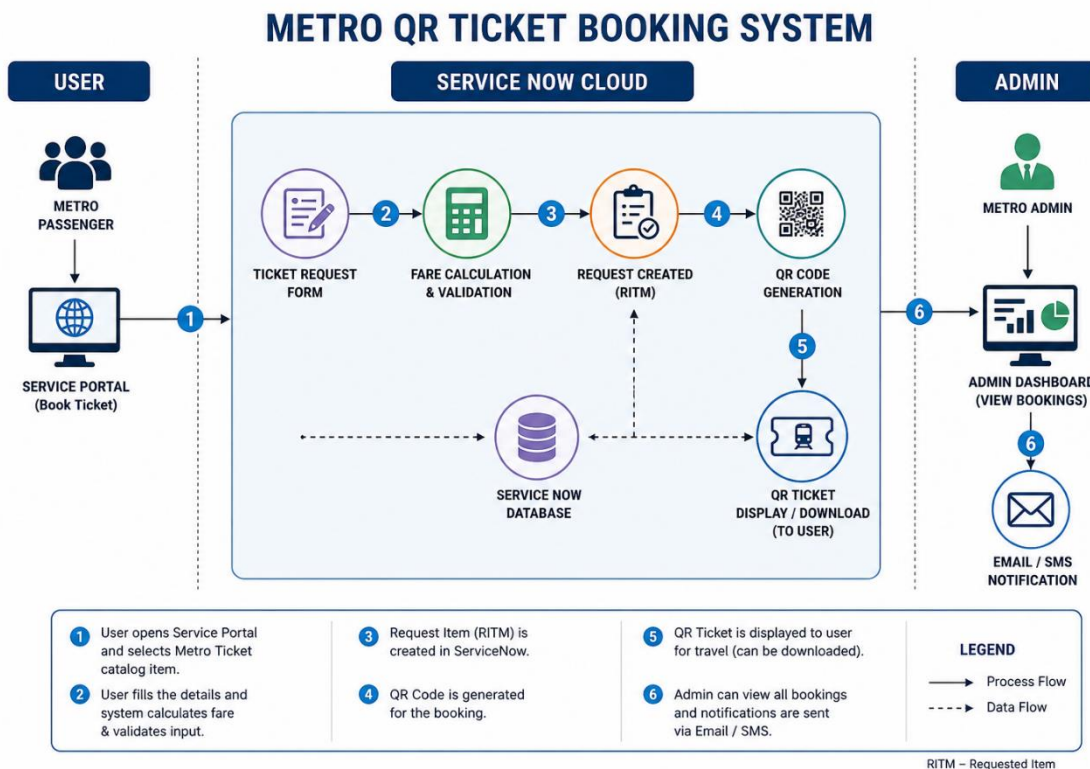


Requirement Analysis Technology Stack (Architecture & Stack)

Date	30 July 2026
Name	Kesireddy Padma
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:



Guidelines:

- ☐ User books the metro ticket through the **ServiceNow Service Portal**.
- ☐ The system validates user inputs and calculates the fare automatically.
- ☐ Ticket booking requests are processed using **Flow Designer**.
- ☐ QR Code is generated after successful request creation.
- ☐ All booking details are securely stored in the **ServiceNow Database**.
- ☐ The application is deployed on the **ServiceNow Cloud Platform**.
- ☐ External QR Code API is used for generating QR codes.
- ☐ Role-based access ensures secure access for **Users** and **Admins**.
- ☐ Admin can view and manage all metro ticket bookings.

Table-1 : Components & Technologies:

S. No	Component	Description	Technology
1.	User Interface	User books metro tickets through Service Portal	ServiceNow Service Portal, HTML, CSS, JavaScript
2.	Application Logic-1	Ticket booking, fare calculation, form validation	Catalog Client Scripts, UI Policies
3.	Application Logic-2	QR Code generation after booking	ServiceNow Widget, JavaScript
4.	Application Logic-3	Ticket approval and request processing	Flow Designer
5.	Database	Stores metro stations, fare details, ticket requests	ServiceNow Tables (Metro Station's Details, Metro Database Table)
6.	Cloud Database	Stores application data in cloud	ServiceNow Cloud Database
7.	File Storage	QR Images and attachments	ServiceNow Attachments
8.	External API-1	QR Code Generation API	QR Server API (api.qrserver.com).
9.	External API-2	Notification Service	Email Notifications / SMS (Optional Twilio) (Optional)
10.	Infrastructure (Server / Cloud)	Entire application deployed on ServiceNow Cloud Platform	ServiceNow SaaS Cloud

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	JavaScript libraries for client-side functionality	HTML, CSS, JavaScript
2.	Security Implementations	Role-based access, ACLs, encrypted HTTPS	ServiceNow ACL, HTTPS, User Roles
3.	Scalable Architecture	Three-tier cloud architecture supporting multiple users	ServiceNow Cloud Platform
4.	Availability	High availability with ServiceNow cloud infrastructure	ServiceNow Data Centers
5.	Performance	Fast request processing using optimized workflows	Flow Designer, ServiceNow Platform

References:

<https://developer.servicenow.com/>

<https://docs.servicenow.com/>

<https://goqr.me/api/>

